Notes: Melitz (Econometrica, 2003)

The Impact of Trade on Intra-Industry Reallocations and Aggregate Industry Productivity

Contents

1 Big picture

- **Question.** How does trade change industry structure when firms differ in productivity, and can trade raise welfare through within-industry reallocations rather than only through more varieties?
- **Setting.** The paper builds a dynamic industry model of monopolistic competition with heterogeneous firms, sunk entry, costly exporting, and endogenous exit. It studies stationary equilibria in autarky and in an open economy with symmetric countries.
- **Core challenge.** Standard representative-firm trade models cannot explain why exporters are systematically more productive than nonexporters, why the least productive firms exit when trade expands, or why industry productivity rises even if firm-level technology does not change.
- **Main conceptual contribution.** The paper combines the Krugman (1980) love-of-variety trade model with Hopenhayn-style firm selection. The resulting model stays tractable because the entire productivity distribution matters for aggregates only through one sufficient statistic, the average productivity level $\tilde{\varphi}$.
- **Main mechanisms.**
  - Firms pay a sunk domestic entry cost before learning productivity.
  - Only sufficiently productive firms survive after drawing productivity.
  - Exporting requires an additional fixed cost and iceberg trade costs, so only an even more productive subset exports.
  - Trade reallocates output and profits toward the most productive firms and forces the least productive firms to exit.
- **Main equilibrium objects.** The key cutoffs are the domestic survival cutoff φ^* and the exporter cutoff φ_x^*. The key aggregate objects are the mass of firms M, the mass of varieties available to consumers M_t, and the weighted average productivity $\tilde{\varphi}$.
- **Main closed-economy result.** In autarky there is a unique cutoff φ^* pinned down by a zero-cutoff-profit condition and a free-entry condition. Country size affects the number of firms and varieties, but not firm-level outcomes such as the cutoff or average profit.
- **Main open-economy result.** Opening to trade raises the domestic productivity cutoff, creates a higher export cutoff, lowers the number of domestic producers, reallocates market shares toward better firms, increases average productivity, and raises welfare.
- **Distributional result across firms.** All firms lose domestic market share. Exporters gain total sales because foreign revenues more than offset part of the domestic loss. But only the most productive exporters gain profits; some exporters still lose profits because they must pay fixed export costs.
- **Key comparative-static result.** More trading partners, lower iceberg costs, or lower fixed export costs all raise welfare. The first two channels also increase the market share of incumbent exporters; lower fixed export costs mainly work by creating new exporters.

- **Economic takeaway.** Trade does not simply add varieties. It changes which firms survive, which firms export, and how output is allocated inside an industry. Those reallocations create an aggregate productivity gain that becomes part of the welfare gain from trade.

2 Model environment and key objects

2.1 Empirical facts the model is built to explain

The paper starts from three stylized facts emphasized by plant- and firm-level data:

- productivity differs persistently across firms within narrowly defined industries;
- exporters are systematically more productive than nonexporters;
- trade liberalization is associated with reallocation of market shares toward better firms and with industry productivity growth.

Interpretation. The model is designed to produce these patterns endogenously. It is not a reduced-form empirical paper; its discipline comes from matching the direction and logic of these widely documented facts.

2.2 Demand

Preferences over a continuum of varieties $\omega \in \Omega$ are CES:

$$U = \left[\int_{\omega\in\Omega} q(\omega)^{\rho}\, d\omega\right]^{1/\rho}, \qquad 0 < \rho < 1, \qquad \sigma = \frac{1}{1-\rho} > 1. \tag{1}$$

The associated ideal price index is

$$P = \left[\int_{\omega\in\Omega} p(\omega)^{1-\sigma}\, d\omega\right]^{1/(1-\sigma)}. \tag{2}$$

Demand for a single variety and its revenue are

$$q(\omega) = Q\left[\frac{p(\omega)}{P}\right]^{-\sigma}, \tag{3}$$

$$r(\omega) = R\left[\frac{p(\omega)}{P}\right]^{1-\sigma}, \tag{4}$$

where Q is aggregate consumption and $R = PQ$ is aggregate expenditure.

Interpretation. Every firm faces a constant-elasticity demand curve. That is what delivers constant markups and keeps the model analytically tractable.

2.3 Production and firm heterogeneity

A firm with productivity φ uses labor according to

$$\ell = f + \frac{q}{\varphi}, \tag{5}$$

where f is a fixed operating cost and $1/\varphi$ is marginal labor input. The wage is normalized to one. Because firms face CES demand, the pricing rule is

$$p(\varphi) = \frac{1}{\rho\varphi}. \tag{6}$$

Profits are

$$\pi(\varphi) = \frac{r(\varphi)}{\sigma} - f, \tag{7}$$

and revenue can be written as

$$r(\varphi) = R(P\rho\varphi)^{\sigma-1}. \tag{8}$$

Relative output and relative revenue for two firms satisfy

$$\frac{q(\varphi_1)}{q(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma}, \qquad \frac{r(\varphi_1)}{r(\varphi_2)} = \left(\frac{\varphi_1}{\varphi_2}\right)^{\sigma-1}. \tag{9}$$

Interpretation. More productive firms charge lower prices, sell more, and earn higher profits. The model therefore delivers cross-firm size dispersion immediately once firms draw different productivity levels.

2.4 Average productivity as a sufficient statistic

Let M be the mass of domestic producers and $\mu(\varphi)$ the distribution of productivity among surviving firms. The key sufficient statistic is

$$\tilde{\varphi} = \left[\int_0^\infty \varphi^{\sigma-1}\mu(\varphi)\, d\varphi\right]^{1/(\sigma-1)}. \tag{10}$$

Using this object, aggregates can be written exactly as if all firms had the same productivity $\tilde{\varphi}$:

$$P = M^{-1/(\sigma-1)}\, p(\tilde{\varphi}) = \frac{M^{-1/(\sigma-1)}}{\rho\tilde{\varphi}}, \tag{11}$$

$$R = M\, r(\tilde{\varphi}), \tag{12}$$

$$H = M\, \pi(\tilde{\varphi}), \tag{13}$$

where H is aggregate profit.

Interpretation. This is the paper's big tractability result. Heterogeneity matters for equilibrium, but for aggregate outcomes the whole distribution can be summarized by one weighted productivity average.

2.5 Entry, exit, and the domestic cutoff

Potential entrants pay a sunk cost f_e before learning their productivity. After entry, they draw φ from a distribution $g(\varphi)$ with cdf $G(\varphi)$. Each incumbent then faces a constant death shock with probability δ every period.

With no time discounting other than death, the value of a firm is

$$v(\varphi) = \max\left\{0, \frac{\pi(\varphi)}{\delta}\right\}. \tag{14}$$

The domestic cutoff φ^* is defined by

$$\pi(\varphi^*) = 0. \tag{15}$$

Firms with $\varphi < \varphi^*$ exit immediately and never produce. The distribution of surviving firms is therefore

$$\mu(\varphi) = \begin{cases} \dfrac{g(\varphi)}{1 - G(\varphi^*)}, & \varphi \geq \varphi^*, \\ 0, & \varphi < \varphi^*. \end{cases} \tag{16}$$

The probability of successful entry is

$$p_{in} = 1 - G(\varphi^*). \tag{17}$$

Interpretation. Selection is endogenous: the cutoff determines which entrants survive, and because only surviving firms are used to compute $\tilde{\varphi}$, it also determines aggregate productivity.

2.6 Exporting technology and the exporter cutoff

Serving a foreign market requires

- an iceberg cost $\tau > 1$, so τ units must be shipped for one to arrive, and
- a fixed export cost f_x per destination market per period (equivalently, the amortized version of a sunk export-entry cost f_{ex}, with $f_x = \delta f_{ex}$).

With symmetric countries, the domestic and export revenue of a firm are

$$r_d(\varphi) = R(P\rho\varphi)^{\sigma-1}, \qquad r_x(\varphi) = \tau^{1-\sigma} r_d(\varphi). \tag{18}$$

Export profit per destination is

$$\pi_x(\varphi) = \frac{r_x(\varphi)}{\sigma} - f_x. \tag{19}$$

The exporter cutoff φ_x^* is defined by $\pi_x(\varphi_x^*) = 0$. Using the domestic cutoff condition $\pi_d(\varphi^*) = 0$, it satisfies

$$\varphi_x^* = \varphi^* \, \tau \left(\frac{f_x}{f}\right)^{1/(\sigma-1)}. \tag{20}$$

Interpretation. Exporting is harder than surviving domestically. The exporter cutoff is therefore above the domestic cutoff whenever trade costs are high enough to partition firms into exporters and nonexporters.

2.7 Varieties consumed and welfare

Let $M_x = p_x M$ be the mass of domestic exporters, where

$$p_x = \frac{1 - G(\varphi_x^*)}{1 - G(\varphi^*)} \tag{21}$$

is the fraction of surviving firms that export.

Consumers in each country have access to

$$M_t = M + nM_x = (1 + np_x)M \tag{22}$$

varieties, where n is the number of foreign partner countries.

The open-economy productivity index relevant for consumer prices combines domestic firms and foreign exporters:

$$\tilde{\varphi}_t = \left[\frac{M\tilde{\varphi}^{\sigma-1} + nM_x(\tau^{-1}\tilde{\varphi}_x)^{\sigma-1}}{M_t}\right]^{1/(\sigma-1)}. \tag{23}$$

Welfare per worker is

$$W = P^{-1} = M_t^{1/(\sigma-1)}\rho\tilde{\varphi}_t. \tag{24}$$

Interpretation. Welfare depends on two margins: how many varieties consumers can access and how productive the firms supplying those varieties are. The paper's novelty is that trade raises welfare through the second margin as well.

3 Model and equilibrium (all equations + interpretation)

3.1 Closed-economy equilibrium: zero-cutoff-profit and free entry

Because $\pi(\varphi^*) = 0$, the cutoff firm's revenue satisfies

$$r(\varphi^*) = \sigma f. \tag{25}$$

Using the relative revenue equation, average profit can be written as

$$\bar{\pi} = \pi(\tilde{\varphi}) = f\,k(\varphi^*), \qquad k(\varphi^*) = \left[\frac{\tilde{\varphi}(\varphi^*)}{\varphi^*}\right]^{\sigma-1} - 1. \tag{26}$$

This is the **zero-cutoff-profit (ZCP)** condition.

The **free-entry (FE)** condition equates the expected value of entry to the sunk entry cost:

$$v_e = p_{in}\frac{\bar{\pi}}{\delta} - f_e = 0 \quad\Longrightarrow\quad \bar{\pi} = \frac{\delta f_e}{1 - G(\varphi^*)}. \tag{27}$$

Interpretation. ZCP says the marginal surviving firm breaks even. FE says entrants are exactly indifferent between entering and not entering. Their intersection pins down a unique φ^* and $\bar{\pi}$.

3.2 Closed-economy stationarity, number of firms, and welfare

In a stationary equilibrium, successful entrants must replace firms hit by the death shock:

$$p_{in} M_e = \delta M. \tag{28}$$

Using labor-market clearing, aggregate revenue equals the labor endowment L. This implies

$$M = \frac{L}{\sigma(\bar{\pi} + f)}. \tag{29}$$

A useful expression for welfare in autarky is

$$W_a = \rho \left(\frac{L}{\sigma f} \right)^{1/(\sigma-1)} \varphi_a^*. \tag{30}$$

Interpretation. In the closed economy, country size changes the number of firms and therefore variety, but it does not change the domestic productivity cutoff. That is why this model differs from representative-firm variety models only once trade costs interact with firm heterogeneity.

3.3 Open-economy firm problem

Domestic and export profits are

$$\pi_d(\varphi) = \frac{r_d(\varphi)}{\sigma} - f, \qquad \pi_x(\varphi) = \frac{r_x(\varphi)}{\sigma} - f_x. \tag{31}$$

Total revenue for a domestic firm is

$$r(\varphi) = \begin{cases} r_d(\varphi), & \varphi < \varphi_x^*, \\ r_d(\varphi) + n r_x(\varphi) = \left(1 + n\tau^{1-\sigma}\right) r_d(\varphi), & \varphi \geq \varphi_x^*. \end{cases} \tag{32}$$

Likewise, total profits are

$$\pi(\varphi) = \pi_d(\varphi) + \max\{0, n\pi_x(\varphi)\}. \tag{33}$$

Interpretation. The domestic market remains open only to firms above φ^*. Export markets are open only to firms above φ_x^*. The model therefore partitions firms into three groups: immediate exiters, domestic-only firms, and exporters.

3.4 Open-economy equilibrium conditions

The ZCP condition now has two pieces because some firms export:

$$\bar{\pi} = \pi_d(\tilde{\varphi}) + p_x n \pi_x(\tilde{\varphi}_x) = f\, k(\varphi^*) + p_x n f_x\, k(\varphi_x^*). \tag{34}$$

The FE condition is unchanged:

$$\bar{\pi} = \frac{\delta f_e}{1 - G(\varphi^*)}. \tag{35}$$

Average firm revenue becomes

$$\bar{r} = r_d(\tilde{\varphi}) + p_x n r_x(\tilde{\varphi}_x) = \sigma\big(\bar{\pi} + f + p_x n f_x\big), \tag{36}$$

and the mass of domestic producers is

$$M = \frac{L}{\sigma(\bar{\pi} + f + p_x n f_x)}. \tag{37}$$

Interpretation. The FE curve is the same as in autarky, but the ZCP curve shifts upward when trade becomes available. The domestic productivity cutoff must therefore rise.

3.5 What opening to trade does relative to autarky

Let φ_a^* be the autarky cutoff. Opening to trade implies

$$\varphi^* > \varphi_a^*. \tag{38}$$

That single inequality implies several results:

- the least productive firms that survived in autarky now exit;
- average productivity rises because the lower tail of the productivity distribution is truncated more aggressively;
- average profit rises;
- the mass of domestic producers M falls;
- welfare rises.

The paper also shows that domestic firms typically offer consumers more total varieties because foreign exporters enter, even though the number of domestic firms falls. But even if variety were to fall, welfare can still rise because the productivity effect dominates.

Interpretation. This is the central result of the paper: trade changes the selection rule inside the industry, and that selection effect itself is welfare-improving.

3.6 Revenue and profit reallocation across firms

For any firm productive enough to remain active,

$$r_d(\varphi) < r_a(\varphi), \tag{39}$$

so every surviving domestic firm loses some domestic sales after trade opens.

For exporters,

$$r_a(\varphi) < r_d(\varphi) + n r_x(\varphi), \qquad \varphi \geq \varphi_x^*, \tag{40}$$

so foreign sales more than offset the domestic loss.

Profit changes are more subtle. For exporters,

$$\Delta\pi(\varphi) = \pi(\varphi) - \pi_a(\varphi) = \frac{1}{\sigma}\Big(r_d(\varphi) + nr_x(\varphi) - r_a(\varphi)\Big) - nf_x, \tag{41}$$

which is increasing in φ.

Interpretation. Exporters are not all winners. Less productive exporters gain revenue but may still lose profit because they must pay fixed export costs. Only the most productive exporters gain profits, and the profit gain rises with productivity.

3.7 Why trade forces the least productive firms to exit

A key conceptual point in the paper is that the selection effect does **not** come from tougher import competition in the usual markup sense. Under CES demand, the elasticity of demand and hence the markup stay fixed.

Instead, the mechanism is:

- high-productivity firms now have profitable export opportunities;
- entrants respond to the possibility of drawing a high productivity and becoming an exporter;
- exporting and entry both absorb labor resources;
- with total resources fixed, the least productive firms can no longer cover fixed costs and are forced out.

Interpretation. The model's selection logic works through resource reallocation and fixed export costs, not through an endogenous rise in markups or tougher price competition.

3.8 Trade liberalization once the economy is already open

The paper studies three liberalization channels.

1. More partner countries ($n \uparrow$).

- φ^* rises.
- φ_x^* also rises.
- All firms lose domestic sales.
- Exporters expand total sales and the most productive exporters increase profits.
- Welfare rises.

2. Lower iceberg trade cost ($\tau \downarrow$).

- φ^* rises.
- φ_x^* falls.
- The least productive firms exit and some domestic-only firms become exporters.

- Welfare rises.

3. Lower fixed export cost ($f_x \downarrow$).

- φ^* rises.
- φ_x^* falls.
- New exporters enter foreign markets.
- Existing exporters do *not* necessarily gain market share or profit; the main effect runs through entry of new exporters.
- Welfare rises.

Interpretation. The broad selection logic is robust to the way trade exposure increases, but the exact distribution of gains across exporters differs depending on whether liberalization lowers variable or fixed trade costs.

4 What makes the results credible (selection, assumptions, and limits)

4.1 Why selection is truly endogenous

Selection comes from two sunk-cost decisions made at different moments:

- firms pay f_e before learning productivity, so some entrants later discover they are too unproductive to operate;
- firms pay export costs only after learning productivity, so exporting becomes a second, more selective decision.

Interpretation. This timing is what generates both the domestic survival cutoff and the exporter cutoff. It is the core reason the model can rationalize observed exporter premia.

4.2 Why the model remains analytically tractable

The sufficient statistic $\tilde{\varphi}$ is what makes the model easy to solve despite heterogeneous firms.

This works because:

- CES demand gives every firm the same constant markup;
- aggregate demand depends on the price index only through a fixed elasticity;
- revenue ratios across firms depend only on relative productivity.

Interpretation. The tractability is not accidental; it rests on the specific combination of CES demand and monopolistic competition. Without that structure, the model would be much harder to solve analytically.

4.3 Why welfare rises in the model

The paper shows that welfare can be written as a monotone function of the domestic cutoff φ^*. Since opening to trade or liberalizing trade raises φ^*, welfare rises.

Interpretation. The welfare gain is not only a variety gain. It also comes from improving the productivity composition of the firms that remain active.

4.4 Assumptions doing the heavy lifting

Several assumptions are central:

- **CES demand.** Markups are fixed, so there is no endogenous pro-competitive markup channel.
- **Symmetric countries.** The paper avoids wage and country-size asymmetries in the main analysis.
- **Stationary equilibrium.** The model focuses on long-run comparative statics, not transition dynamics.
- **Simplified exit dynamics.** Firms face an iid death shock rather than a full productivity-evolution process.

Interpretation. These assumptions are precisely what let the model isolate the selection channel cleanly. They also mark the limits of the paper.

4.5 What the model can and cannot explain

The model **can** explain:

- why exporters are more productive than nonexporters;
- why trade pushes least productive firms out;
- why industry productivity rises after trade expands even if firm technology does not improve.

The model **cannot** explain:

- endogenous markup changes;
- transition dynamics after a policy shock;
- firm-level investment in innovation or product quality;
- rich country asymmetries with endogenous relative wages.

Interpretation. The paper is strongest as a clean benchmark for the selection channel. Later work can add richer competitive or dynamic features on top of it.

4.6 Relation to the literature

- Relative to **Krugman (1980)**, the paper adds firm heterogeneity and endogenous selection.

- Relative to **Hopenhayn (1992a, 1992b)**, it embeds selection into a monopolistic-competition trade framework.
- Relative to **Bernard, Eaton, Jensen, and Kortum (2000)**, it keeps markups exogenous but delivers transparent analytical comparative statics.
- The paper is also a foundation for later work such as **Helpman, Melitz, and Yeaple (2004)** on export-versus-FDI choices.

Interpretation. The paper's place in the literature is to provide the canonical trade model with heterogeneous firms and endogenous export selection.

5 Figures and key comparative-static results

5.1 Figure 1: equilibrium cutoff and average profit

Read:

- The downward-sloping curve is the zero-cutoff-profit schedule: as the cutoff φ^* rises, the average productivity advantage of surviving firms shrinks relative to the cutoff, so average profit falls.
- The upward-sloping curve is the free-entry schedule: a higher cutoff means successful entry is rarer, so surviving firms must earn more on average to justify paying the sunk entry cost.
- The intersection determines the unique equilibrium pair $(\varphi^*, \bar{\pi})$.

Economic message. The model has a clean, one-dimensional selection structure. Everything in autarky starts from the single equilibrium cutoff where marginal survivors break even and entrants are just indifferent.

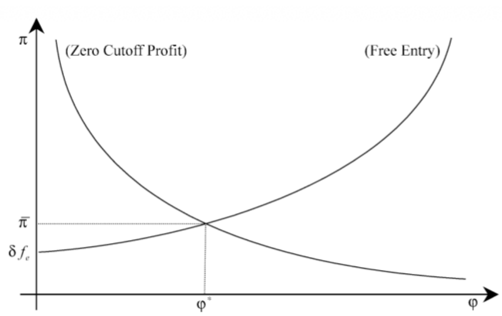

FIGURE 1.—Determination of the equilibrium cutoff φ^* and average profit $\bar{\pi}$.

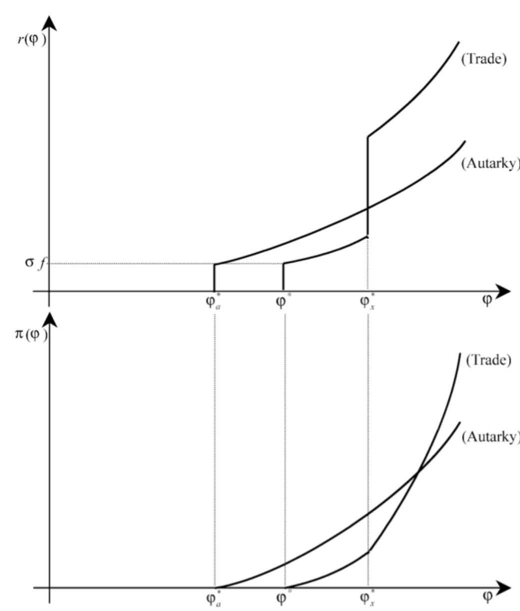

FIGURE 2.—The reallocation of market shares and profits.

5.2 Figure 2: trade reallocates market shares and profits

Read:

- The top panel plots revenue against productivity. The autarky cutoff φ_a^* lies to the left of the open-economy domestic cutoff φ^*, which itself lies to the left of the exporter cutoff φ_x^*.
- Moving from autarky to trade shifts down domestic revenue for every surviving firm, but exporters get a jump in total revenue because they add foreign sales.
- The bottom panel plots profits. Domestic-only firms lose profits, low-productivity exporters may still lose profits, and only the most productive exporters gain profits.

Economic message. Trade benefits are uneven. It is not true that every exporting firm wins. What trade does is tilt the whole industry toward its most productive firms.

5.3 Opening to trade: the main propositions

Read:

- Opening to trade shifts the ZCP schedule upward relative to autarky, so the domestic cutoff rises from φ_a^* to φ^*.
- The number of domestic firms falls, but consumers gain access to foreign exporters, so available varieties usually rise.
- Aggregate productivity rises because both exit and export selection push market shares toward more productive firms.
- Welfare rises even if variety does not rise enough on its own, because the productivity-composition effect is positive.

Economic message. The model's central qualitative result is that trade can raise welfare by improving *who produces*, not only by expanding *how many varieties* are sold.

5.4 Trade liberalization: more partners, lower trade costs, and lower export fixed costs

Read:

- More trading partners ($n \uparrow$) raise the domestic cutoff and intensify reallocation toward exporters.
- Lower variable trade costs ($\tau \downarrow$) both raise the domestic cutoff and lower the exporter cutoff, so more firms export and the least productive firms exit.
- Lower fixed export costs ($f_x \downarrow$) also lower the exporter cutoff, but the main effect is entry of new exporters rather than extra rents for incumbent exporters.

Economic message. The selection logic is robust, but the exact pattern of gains differs depending on whether liberalization lowers fixed or variable costs of serving foreign markets.

6 Short summary

- The paper builds a dynamic trade model with heterogeneous firms, sunk entry, fixed export costs, and endogenous exit.
- A domestic productivity cutoff determines which firms survive; a higher export cutoff determines which survivors export.
- Trade raises the domestic cutoff, makes average productivity higher, and reallocates market shares toward more productive firms.
- All firms lose domestic market share, exporters gain total sales, but only the most productive exporters gain profits.
- Welfare rises not only because consumers access more varieties but also because trade improves the productivity composition of active firms.
- More trading partners, lower iceberg costs, and lower fixed export costs all raise welfare, though they affect incumbent exporters differently.
- The model's main limitation is the CES environment with fixed markups, which shuts down a pro-competitive markup channel and keeps the focus on selection.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that trade changes industry performance through an endogenous selection mechanism. Once firms differ in productivity and exporting is costly, trade induces the most productive firms to export, forces the least productive firms to exit, and reallocates output and profits inside the industry. Those reallocations raise aggregate productivity and therefore welfare.

7.2 Economic intuition

Trade does not treat all firms equally. The most productive firms can afford to pay export costs and spread fixed costs over larger sales. The least productive firms cannot survive once scarce resources are pulled toward exporters and entrants. The industry becomes leaner, more export-oriented, and more productive on average.

7.3 Takeaway

The paper's lasting message is that the gains from trade are partly *selection gains*. Trade matters not only because it expands markets, but because it changes *which firms produce* and *which firms grow*. That idea is now central to modern trade theory.